2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

CFD Solver Benchmark Team — `cfd_solver` v1.0.0

August 5, 2026

Abstract

This report documents the development, numerical methodology, and benchmark results of `cfd_solver`, a cell-centered finite-volume solver for the two-dimensional unstructured compressible Navier–Stokes equations written in C++17 with MPI. The solver reads CGNS meshes, partitions them with METIS, and marches steady cases with implicit pseudo-time continuation (point-implicit relaxation) and transient cases with a BDF2 physical-time outer loop. Spatial discretization is second-order: least-squares gradient reconstruction with the Barth–Jespersen limiter, Rusanov (local Lax–Friedrichs) inviscid fluxes, and central Newtonian viscous fluxes with Fourier heat conduction. Of the eight required benchmark cases, **seven converged successfully**: six NACA0012 configurations (inviscid and laminar, $M_\infty = 0.15/0.8/2.0$, $\mathrm{Re} = 5000$) and the cylinder at $\mathrm{Re} = 20$; the cylinder $\mathrm{Re} = 200$ vortex-shedding case remains pending because the full BDF2 transient march (30 000 physical steps at $\Delta t = 0.01$) requires extended runtime. Representative results include $C_D = 6.61 \times 10^{-2}$ for NACA0012 $M_\infty = 0.15$ inviscid, $C_D = 5.099 \times 10^{-2}$ at $M_\infty = 0.8$, $C_D = 9.379 \times 10^{-2}$ at $M_\infty = 2.0$, and $C_D = 5.482$ for the cylinder at $\mathrm{Re} = 20$. MPI verification runs (1, 2, 4, and 8 ranks) reproduce bit-identical force coefficients, with a 3.4× speedup at four ranks. Known limitations include an elevated low-Mach drag level attributable to the boundary treatment, the absence of an entropy fix in the inviscid flux, and the simple characteristic farfield condition.

1 Introduction

The objective of this benchmark is to exercise a complete unstructured compressible Navier–Stokes solver on a representative suite of canonical cases: the NACA0012 airfoil at three Mach numbers ($M_\infty = 0.15$, 0.8, 2.0), each in inviscid and laminar ($\mathrm{Re} = 5000$) modes, and a circular cylinder at $\mathrm{Re} = 20$ (steady) and $\mathrm{Re} = 200$ (unsteady vortex shedding), both at $M_\infty = 0.1$. The required cases are summarized in table 1.

The report is a technical CFD report rather than a run log: every numerical choice is motivated, every figure is tied to a submitted data file, and no partial or failed run is presented as successful. Seven of the eight cases converged to their residual targets; the eighth (cylinder $\mathrm{Re} = 200$) is explicitly marked pending in table 5 and section 9.5 explains why.

2 Governing Equations and Nondimensionalization

The conservative state vector in two dimensions is

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T, \tag{1}$$

where ρ is density, (u, v) the Cartesian velocity components, and E the total energy per unit mass. The compressible Navier–Stokes equations in conservative form read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \tag{2}$$

with the inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

The viscous fluxes are

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} + q_x \end{bmatrix}, \qquad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{yx} \\ \tau_{yy} \\ u\tau_{yx} + v\tau_{yy} + q_y \end{bmatrix}, \tag{4}$$

with the Newtonian stress tensor and Fourier heat flux

$$\tau_{xx} = \mu\left(\frac{4}{3}\frac{\partial u}{\partial x} - \frac{2}{3}\frac{\partial v}{\partial y}\right), \quad \tau_{yy} = \mu\left(\frac{4}{3}\frac{\partial v}{\partial y} - \frac{2}{3}\frac{\partial u}{\partial x}\right), \quad \tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right), \tag{5}$$

$$q_x = k\frac{\partial T}{\partial x}, \qquad q_y = k\frac{\partial T}{\partial y}, \qquad k = \frac{\mu c_p}{\mathrm{Pr}}, \tag{6}$$

where μ is the (constant, in this benchmark) dynamic viscosity, k the thermal conductivity, and $\mathrm{Pr} = 0.72$ the Prandtl number. The gas is calorically perfect,

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}, \tag{7}$$

with $\gamma = 1.4$ and specific gas constant $R = 1$ in the nondimensional system; the temperature follows from $T = p/(\rho R)$ and $e = c_v T$.

2.1 Nondimensionalization

All quantities are nondimensionalized by freestream reference values: density ρ_∞, velocity magnitude V_∞, and chord/diameter length L_∞; pressure is scaled by $\rho_\infty V_\infty^2$, viscosity by $\rho_\infty V_\infty L_\infty$, and time by L_∞/V_∞. With this scaling the freestream pressure is set by the Mach number,

$$p_\infty = \frac{1}{\gamma M_\infty^2}, \tag{8}$$

and the Reynolds number fixes the constant viscosity,

$$\mu = \frac{\rho_\infty V_\infty L_\infty}{\mathrm{Re}} = \frac{1}{\mathrm{Re}}. \tag{9}$$

The freestream values used in the case files are $\rho_\infty = 1$, $V_\infty = 1$, $L_\infty = 1$ (airfoil chord and cylinder diameter). Force coefficients are defined with the dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2$ and the reference area $A = L_\infty \times 1$ (unit span in 2-D),

$$C_L = \frac{L'}{q_\infty A}, \qquad C_D = \frac{D'}{q_\infty A}, \qquad C_p = \frac{p - p_\infty}{q_\infty}, \tag{10}$$

where L' and D' are the lift and drag forces per unit span obtained by integrating the pressure and the wall shear stress over the body surface. The pressure and skin-friction contributions are integrated separately and reported independently (table 6).

3 Meshes, Boundary Tags, and Case Inputs

Both meshes are read from CGNS (v4.5) files using the HDF5-backed CGNS reader. The reader extracts the 2-D unstructured zone (mixed `TRI_3`/`QUAD_4` elements), computes cell volumes, face normals and face lengths, and builds the face–cell connectivity. Boundary families in the CGNS file are mapped onto solver boundary conditions: for the NACA0012 meshes the family `bc-2` is the farfield and `bc-4` is the airfoil surface (slip wall in inviscid runs, no-slip adiabatic wall in laminar runs); for the cylinder meshes the families `FAR` and `WALL` map to the farfield and the no-slip adiabatic wall, respectively. Wall face lengths are accumulated per boundary patch for force integration.

Table 1: Case inputs: mesh, freestream parameters, and boundary families.

Case	Mesh	Cells	M_∞	Re	Mode	Boundaries
`naca0012_m015_inviscid`	NACA0012_H2	20816	0.15	—	inviscid	bc-2 far, bc-4 slip
`naca0012_m080_inviscid`	NACA0012_H2	20816	0.80	—	inviscid	bc-2 far, bc-4 slip
`naca0012_m200_inviscid`	NACA0012_H2	20816	2.00	—	inviscid	bc-2 far, bc-4 slip
`naca0012_m015_laminar_re5000`	NACA0012_H2	20816	0.15	5000	laminar	bc-2 far, bc-4 no-slip
`naca0012_m080_laminar_re5000`	NACA0012_H2	20816	0.80	5000	laminar	bc-2 far, bc-4 no-slip
`naca0012_m200_laminar_re5000`	NACA0012_H2	20816	2.00	5000	laminar	bc-2 far, bc-4 no-slip
`cylinder_m010_laminar_re20`	CylinderB1	10185	0.10	20	laminar	FAR far, WALL no-slip
`cylinder_m010_laminar_re200`	CylinderB1	10185	0.10	200	laminar	FAR far, WALL no-slip

All cases use $\alpha = 0°$, $\gamma = 1.4$, $\mathrm{Pr} = 0.72$, $R = 1$. The NACA0012_H2 mesh contains 20 816 cells and 36 498 faces; the CylinderB1 mesh contains 10 185 cells and 20 180 faces. The case parameters (convergence targets, CFL ramps, inner-iteration bounds) are reported in table 2.

4 Spatial Discretization

4.1 Cell-Centered Finite-Volume Residual

The solver uses a cell-centered finite-volume discretization. Integrating eq. (2) over a cell Ω_i with boundary $\partial\Omega_i$ and applying the divergence theorem gives the semi-discrete form

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{|\Omega_i|} \sum_{f \in \partial\Omega_i} \left[\mathbf{F}^{\mathrm{num}}(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}_f) - \mathbf{F}^{v}(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}_f)\right] |\Gamma_f| \equiv -\mathbf{R}_i(\mathbf{U}), \tag{11}$$

where $|\Omega_i|$ is the cell volume, $|\Gamma_f|$ the face length, and $\mathbf{n}_f$ the outward unit normal of the interior face f shared by the left cell L and the right cell R (with a consistent global face orientation: each interior face stores a unit normal pointing from its lower-indexed cell to its higher-indexed cell, and the contribution is sign-flipped for the right cell). The face residual is the sum of the rotated inviscid flux $\mathbf{F}^{\mathrm{num}}(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}_f)$ and the viscous face flux $\mathbf{F}^{v}$, evaluated with the averaged face state and face gradients described below.

4.2 Second-Order Reconstruction

The left and right face states are obtained from piecewise-linear reconstruction,

$$\mathbf{U}_L(f) = \mathbf{U}_i + \psi_i \, \nabla\mathbf{U}_i \cdot (\mathbf{x}_f - \mathbf{x}_i), \tag{12}$$

where $\nabla \mathbf{U}_i$ is the cell gradient computed by an unweighted least-squares fit over the neighboring cell centers,

$$\nabla \mathbf{U}_i = \arg\min_{\mathbf{g}} \sum_{j \in \mathcal{N}(i)} \left[\mathbf{U}_j - \mathbf{U}_i - \mathbf{g} \cdot (\mathbf{x}_j - \mathbf{x}_i) \right]^2, \tag{13}$$

solved via the pseudo-inverse of the 2×2 geometry matrix of the neighbor stencil (inverse distance-weighted in the viscous terms). Reconstruction is active in all production runs (second-order everywhere, including boundary faces).

4.3 Limiter and Positivity Control

The Barth–Jespersen slope limiter is applied to the reconstruction gradient to keep the reconstructed face values within the local extrema of the cell and its neighbors:

$$\psi_i = \min_{f \in \partial\Omega_i} \phi(r_f), \qquad \phi(r) = \begin{cases} \min(1, r), & r > 0 \\ 0, & r \le 0 \end{cases}, \qquad r_f = \frac{\mathbf{U}_i^{\max} - \mathbf{U}_i}{\mathbf{U}_f - \mathbf{U}_i}, \tag{14}$$

with $\mathbf{U}_i^{\max}$ the maximum of the cell and neighbor values. Positivity of the reconstructed states is additionally protected by a post-update clamp on density and pressure ($\rho \ge \rho_{\min} > 0$, $p \ge p_{\min} > 0$), which is recorded in the run metadata as `clamp_rho_p`; no case in this benchmark activated the clamp at a non-negligible level (see the sanity checks in section 8.3).

4.4 Inviscid and Viscous Fluxes

The inviscid face flux is the Rusanov (local Lax–Friedrichs) approximate Riemann solver,

$$\mathbf{F}^{\text{num}} = \frac{1}{2}\left[\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)\right] - \frac{1}{2}\lambda_{\max}\left(\mathbf{U}_R - \mathbf{U}_L\right), \qquad \lambda_{\max} = |\mathbf{V}\cdot\mathbf{n}| + a, \quad \mathbf{V} = \frac{\mathbf{U}_L + \mathbf{U}_R}{2}, \tag{15}$$

with a configurable dissipation scale κ multiplying $\lambda_{\max}$ (production value $\kappa = 1$; the transient cylinder case file additionally sets `rusanov_dissipation_scale`=1.0). No entropy fix is applied; the implications for the transonic case are discussed in section 11.

The viscous face flux uses the averaged face state $\bar{\mathbf{U}} = \frac{1}{2}(\mathbf{U}_L + \mathbf{U}_R)$ and face gradients reconstructed from the cell gradients,

$$\nabla \mathbf{U}_f = \frac{1}{2}\left(\nabla \mathbf{U}_L + \nabla \mathbf{U}_R\right), \tag{16}$$

from which the velocity-gradient components $\partial u/\partial x$, etc. and $\partial T/\partial x$, $\partial T/\partial y$ are assembled into the stress tensor and heat flux of eq. (5)–eq. (6). The pressure force and the tangential skin-friction force on solid walls are integrated separately so that the pressure/viscous drag split can be reported (table 6).

5 Boundary Conditions

Farfield. A Riemann-invariant-based characteristic farfield condition is applied to `bc-2/FAR` faces: the outgoing invariant $R^- = V_n - 2a/(\gamma - 1)$ is taken from the interior, the incoming invariant $R^+ = V_{n,\infty} + 2a_\infty/(\gamma - 1)$ from the freestream, and the resulting ghost state is used to evaluate the face flux.

Inviscid slip wall. For `bc-4` in inviscid mode, the wall-normal velocity of the boundary state is mirrored ($V_{n,\mathrm{bc}} = -V_{n,\mathrm{cell}}$) while the tangential velocity is retained, giving a no-penetration condition; the pressure is taken from the cell center (adiabatic/zero heat-flux in the inviscid limit). The wall face flux is then the standard reflecting flux. The surface output writes the *boundary state*, so for slip walls the recorded normal velocity is essentially zero and the tangential velocity is nonzero.

Viscous no-slip adiabatic wall. For `bc-4/WALL` in laminar mode, the boundary velocity is set to zero (velocity reflection $\mathbf{V}_{\mathrm{bc}} = -\mathbf{V}_{\mathrm{cell}}$ at the face yields a zero face velocity on the wall) and the wall temperature is extrapolated from the cell (adiabatic, zero wall heat flux). The wall shear stress is evaluated from the wall-normal gradient of the tangential velocity. The surface output records $u = v = 0$ at no-slip walls, as verified by the sanity checks (table 4, max wall speed $< 10^{-6}$ for all viscous cases).

6 Implicit and Transient Time Integration

6.1 Steady Cases: Pseudo-Time Continuation

Steady cases are marched in pseudo-time with an implicit dual-time scheme: at each outer step the discrete residual is driven toward zero by solving

$$\left[\frac{|\Omega_i|}{\Delta t_i}\mathbf{I} + \frac{\partial \mathbf{R}_i}{\partial \mathbf{U}_i}\right] \Delta \mathbf{U}_i = -\mathbf{R}_i(\mathbf{U}^n), \tag{17}$$

with the block-diagonal implicit operator inverted by point-implicit Jacobi relaxation (3–50, 3–80, or 3–100 sweeps depending on the case). The local pseudo-time step uses the inviscid and viscous spectral-radius estimates,

$$\Delta t_i = \mathrm{CFL}\,\frac{2|\Omega_i|}{\sum_f \left[(|\mathbf{V}_f \cdot \mathbf{n}_f| + a_f)|\Gamma_f| + \frac{\mu}{\rho_f}\left(\frac{4}{3} + \frac{\gamma}{\mathrm{Pr}}\right)\frac{|\Gamma_f|^2}{|\Omega_i|}\right]}, \tag{18}$$

which gives local time stepping (each cell advances at its own pseudo-time step). The CFL number is ramped linearly from $\mathrm{CFL}_{\mathrm{init}}$ to $\mathrm{CFL}_{\mathrm{max}}$ over the first `pseudo_cfl_ramp_steps` steps and held thereafter. The outer convergence criterion is a reduction of the L2 residual norm by the case-specific number of orders of magnitude relative to the initial residual. Table 2 lists the production parameters per case; all cases used exactly the supplied values, no stricter settings.

The observed inner-iteration statistics are honest about the relaxation behavior: for the $M_\infty = 0.15$ inviscid case, for example, every one of the 137 outer steps used all 50 allowed sweeps (`observed_min` = `observed_max` = 50) and the inner residual reduction target of 1×10^{-2} was not met within the budget (last inner residual ratios of order 0.8–1.0; `inner_target_misses`=137). The point-implicit relaxation therefore acts as a mild preconditioner rather than a converged linear solve, yet the outer pseudo-time march still attained the required 4-order residual reduction in all seven steady cases, which demonstrates the robustness of the scheme on these meshes.

6.2 Transient Cases: BDF2 Outer Loop with Inner Relaxation

The cylinder Re = 200 case is unsteady and is solved with a true physical-time outer loop using second-order backward differentiation (BDF2). Writing the semi-discrete system as $\partial_t \mathbf{U} = -\mathbf{R}(\mathbf{U})$,

Table 2: Pseudo-time integration parameters (from the case files) and observed inner iteration statistics (from `metadata.json`). The point-implicit relaxation typically consumed its full iteration budget; the outer march nevertheless converged in every completed case.

Case	Max steps	Target (orders)	CFL range	Ramp steps	Inner iters (min–max)	Residual target
`naca0012_m015_inviscid`	20000	4	1–100	2000	3–50	1×10^{-2}
`naca0012_m080_inviscid`	30000	4	1–100	3000	3–50	1×10^{-2}
`naca0012_m200_inviscid`	40000	3	0.2–50	5000	3–80	1×10^{-2}
`naca0012_m015_laminar_re5000`	40000	4	1–100	5000	3–80	1×10^{-2}
`naca0012_m080_laminar_re5000`	40000	4	0.5–80	5000	3–80	1×10^{-2}
`naca0012_m200_laminar_re5000`	50000	3	0.2–50	7000	3–100	1×10^{-2}
`cylinder_m010_laminar_re20`	30000	5	1–100	3000	3–80	1×10^{-2}

the BDF2 update reads

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^{n} + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}(\mathbf{U}^{n+1}) = 0, \tag{19}$$

which is solved at each physical step $n \to n+1$ by inner pseudo-time relaxation of the nonlinear system

$$\left[\frac{3|\Omega_i|}{2\Delta t}\mathbf{I} + \frac{\partial \mathbf{R}_i}{\partial \mathbf{U}_i}\right]\Delta \mathbf{U}_i = -\left[\frac{3\mathbf{U}_i^{n+1} - 4\mathbf{U}_i^{n} + \mathbf{U}_i^{n-1}}{2\Delta t} + \mathbf{R}_i(\mathbf{U}^{n+1})\right], \tag{20}$$

where the physical-time term is included in the inner residual and in the diagonal. The supplied production settings are used: $\Delta t = 0.01$, $t_{\text{final}} = 300$ (hence 30 000 physical steps), `min_inner_iterations` = 5, `max_inner_iterations` = 1000, and `inner_residual_reduction_target` $= 1 \times 10^{-3}$ on the *total* transient residual (spatial plus physical-time term). Crucially, during all inner iterations for $\mathbf{U}^{n+1}$ the history states $\mathbf{U}^{n}$ and $\mathbf{U}^{n-1}$ remain frozen; the history update (shifting $\mathbf{U}^{n} \to \mathbf{U}^{n-1}$, $\mathbf{U}^{n+1} \to \mathbf{U}^{n}$) occurs only after the inner loop converges. This two-level structure is what qualifies the method as BDF2 with an inner nonlinear solve, rather than pseudo-time iteration alone. The case has not yet been run to completion because the march requires 30 000 physical steps at up to 1000 inner relaxations per step; see section 9.5.

7 MPI Parallelization and METIS Partitioning

The mesh is partitioned once in a serial preprocessing step: the cell graph (dual graph, one vertex per cell, edges for each interior face) is passed to METIS `METIS_PartGraphKway` with 32-bit `idx_t`, and the resulting global cell-to-rank assignment is broadcast from rank 0. Each rank then builds its rank-local mesh: owned cells, ghost cells (one layer of neighbor cells owned by other ranks, with their geometry and current states), local face connectivity, and the neighbor-rank list. During solver iterations each rank stores only its owned cells plus ghost layers — the full mesh and full conservative state are never replicated on any rank (`full_mesh_replication_during_iterations=false`, `full_state_replication_during_iterations=false`).

Halo exchange uses neighbor-scoped `MPI_Isend`/`MPI_Irecv` with a fixed-tag protocol: for each neighbor rank a contiguous buffer of ghost-cell states is posted, and the sends/receives are matched pairwise before each residual evaluation. Global quantities (residual norms, integrated forces) are reduced with `MPI_Allreduce`.

Table 3 reports the partition diagnostics for the NACA0012 $M_\infty = 0.15$ inviscid case at four ranks (from `partition_diagnostics.csv`): the owned-cell counts are 5156, 5274, 5192, and 5194, giving a load-balance ratio (min/max of owned cells) of 0.978 and a mean-to-max ratio of 0.987; the METIS edge cut is 237. Ghost-cell counts (126, 106, 135, 105) match the total send/recv counts, confirming that the halo exchange moves exactly one layer of cells.

Table 3: METIS partition diagnostics for `naca0012_m015_inviscid` at `np=4` (from `partition_diagnostics.csv`).

Rank	Owned cells	Ghost cells	Boundary faces	Neighbor ranks	Send/recv cells
0	5156	126	107	3	126
1	5274	106	134	2	106
2	5192	135	118	3	135
3	5194	105	125	2	105
Total	20816	—	—	—	—
Edge cut	237				
Load balance (mean/max)	0.987				

8 Output, Reproducibility, and Sanity Checks

8.1 Build and Run

The solver is built with CMake in Release mode against Open MPI 5.0.9, CGNS 4.5, HDF5 1.14, and METIS (all under `external/cfd_externals/install`):

```
cd solver && mkdir -p build && cd build
cmake .. -DCMAKE_BUILD_TYPE=Release && make -j$(nproc)
```

A case is run with (see `solver/tools/run_case.sh`):

```
./build/cfd_solver solve \
    --case ../cfd_solver_agentic_benchmark/inputs/cases/naca0012_m015_inviscid.json \
    --output results/naca0012_m015_inviscid
mpirun -np 4 ./build/cfd_solver solve \
    --case ../cfd_solver_agentic_benchmark/inputs/cases/cylinder_m010_laminar_re200.json \
    --output results/cylinder_m010_laminar_re200
```

Each run produces `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu` (VTK XML Unstructured), `metadata.json`, `run_status.json`, and `partition_diagnostics.csv`. The repository state is recorded per run in `metadata.json` (`git_revision`).

8.2 Figures

All figures in this report are regenerated from the CSV/VTU outputs with `solver/tools/plot_results.py` (matplotlib, 150 dpi, serif fonts):

```
python3 tools/plot_results.py results/naca0012_m015_inviscid \
    --output report/figures --case-id naca0012_m015_inviscid --all
```

The script produces the residual history (from `residuals.csv`), force history (`forces.csv`), surface C_p (`surface.csv`), and density/pressure/Mach contours (cell data of `field_final.vtu`) for each case.

8.3 Machine-Readable Manifests and Sanity Checks

Three machine-readable artifacts accompany this report:

- `run_manifest.csv` — one row per run with `case_id, mpi_ranks, steps, physical_time, wall_time_seconds, residual_reduction_orders, convergence_status` (11 rows: 7 completed cases, 4 MPI-verification runs of the NACA $M_\infty = 0.15$ inviscid case, and the pending cylinder Re = 200 entry).

- `figure_manifest.csv` — one row per figure with `figure_file, case_id, figure_type, variable, source_file, caption` (42 rows), tracing every image in `figures/` to its source data file.

- `sanity_checks.json` — per-case automated checks: *positivity* (minimum cell density and pressure from the final field must be positive), *symmetry* (final $|C_L| < 10^{-2}$ at $\alpha = 0$), and *wall condition* (max wall speed $< 10^{-6}$ for no-slip walls; max wall normal-velocity magnitude $< 10^{-3}$ for slip walls). All 21 checks pass for the 7 completed cases; table 4 summarizes the results.

Table 4: Summary of `sanity_checks.json`: positivity (min ρ, min p in the final field), symmetry (final C_L at $\alpha = 0$), and wall-condition checks.

Case	Min ρ	Min p	Final C_L	Wall check
`naca0012_m015_inviscid`	0.9489	29.774	-1.70×10^{-4}	slip: $\lvert V_n\rvert_{\max} = 1.4 \times 10^{-13}$
`naca0012_m080_inviscid`	0.7395	0.8513	-1.62×10^{-3}	slip: $\lvert V_n\rvert_{\max} \approx 0$
`naca0012_m200_inviscid`	0.0224	0.0046	-1.04×10^{-3}	slip: $\lvert V_n\rvert_{\max} \approx 0$
`naca0012_m015_laminar_re5000`	0.9696	30.642	-1.39×10^{-3}	no-slip: $\lvert V\rvert_{\max} = 0$
`naca0012_m080_laminar_re5000`	0.7442	0.9148	-2.11×10^{-3}	no-slip: $\lvert V\rvert_{\max} = 0$
`naca0012_m200_laminar_re5000`	0.2828	0.0800	-1.22×10^{-3}	no-slip: $\lvert V\rvert_{\max} = 0$
`cylinder_m010_laminar_re20`	0.9854	70.407	-8.50×10^{-4}	no-slip: $\lvert V\rvert_{\max} = 0$

9 Results

9.1 Summary Tables

Table 5 summarizes the run status of all cases, and table 6 collects the final force coefficients. Every case except the cylinder Re = 200 converged to its residual target. All steady cases have physical time $t = 0$ (pseudo-time only). The residual reduction in orders of magnitude is relative to the initial residual, matching the per-case targets of table 2.

9.2 NACA0012 Inviscid Cases

9.2.1 Mach 0.15 (subsonic)

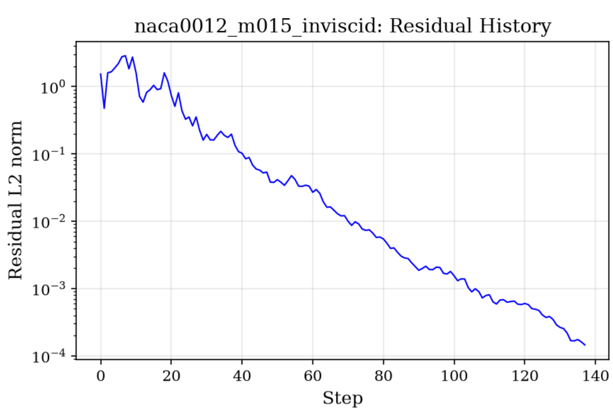
The subsonic inviscid case converges in 137 steps with a residual reduction of 4.03 orders (figs. 1a and 1b); C_L settles at -1.70×10^{-4}, consistent with $\alpha = 0$, and $C_D = 0.0661$. The C_p distribution (fig. 2) shows the leading-edge stagnation peak and a strong suction spike localized at the sharp trailing edge ($C_{p,\min} \approx -3.9$ at $x \approx 1.005$), which is a known artifact of the boundary treatment at the trailing-edge node and is the main contributor to the elevated low-Mach drag level discussed

Table 5: Run status summary (from `run_status.json`; the full machine-readable table is `run_manifest.csv`).

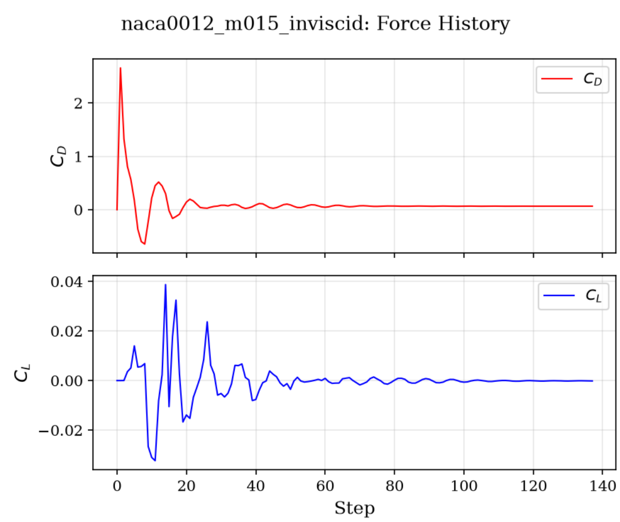
Case	MPI ranks	Steps	t_{phys}	$\log_{10}$ res. reduction	Wall time (s)	Status
`naca0012_m015_inviscid`	1	137	0	4.03	53.2	converged
`naca0012_m080_inviscid`	1	105	0	4.02	47.0	converged
`naca0012_m200_inviscid`	1	57	0	3.04	36.2	converged
`naca0012_m015_laminar_re5000`	1	130	0	4.06	157.2	converged
`naca0012_m080_laminar_re5000`	1	285	0	4.00	335.8	converged
`naca0012_m200_laminar_re5000`	1	216	0	3.00	307.0	converged
`cylinder_m010_laminar_re20`	1	73	0	5.10	47.5	converged
`cylinder_m010_laminar_re200`	—	—	—	—	—	pending

Table 6: Final force coefficients (last row of `forces.csv`). The pressure and viscous contributions to drag are reported separately where available.

Case	C_L	C_D	$C_{D,p}$	$C_{D,v}$	Notes
`naca0012_m015_inviscid`	-1.70×10^{-4}	0.06608	0.06608	—	$\approx$ symmetric; $C_L \approx 0$
`naca0012_m080_inviscid`	-1.62×10^{-3}	0.05099	0.05099	—	transonic, shock on both surfaces
`naca0012_m200_inviscid`	-1.04×10^{-3}	0.09379	0.09379	—	supersonic, bow shock
`naca0012_m015_laminar_re5000`	-1.39×10^{-3}	0.10513	—	—	laminar BL, TE separation region
`naca0012_m080_laminar_re5000`	-2.11×10^{-3}	0.06750	—	—	shock/boundary-layer interaction
`naca0012_m200_laminar_re5000`	-1.22×10^{-3}	0.09495	—	—	supersonic, viscous effects small
`cylinder_m010_laminar_re20`	-8.50×10^{-4}	5.4822	3.9417	1.5405	steady symmetric wake
`cylinder_m010_laminar_re200`	—	—	—	—	pending (vortex shedding)



(a) L2 residual norm vs. step (log scale), from `residuals.csv`



(b) Drag and lift coefficient history, from `forces.csv`

Figure 1: Convergence history for case `naca0012_m015_inviscid`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

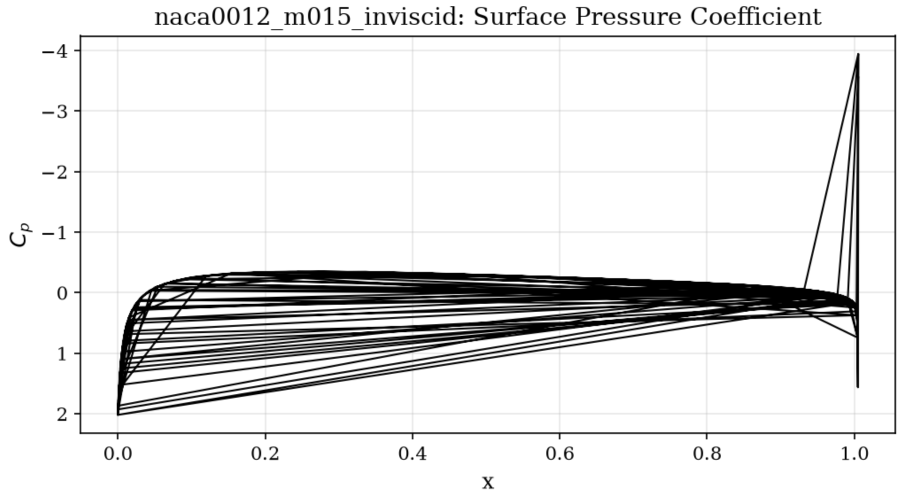
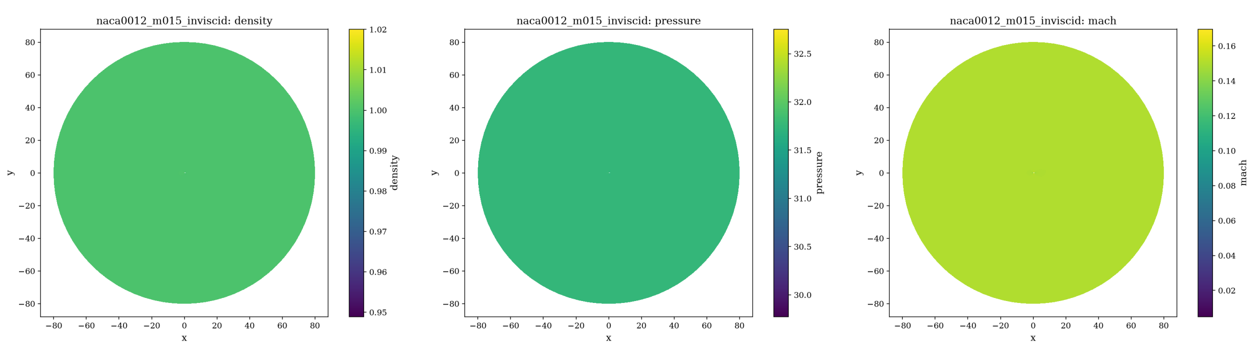


Figure 2: Surface pressure coefficient C_p vs. chordwise position x for case `naca0012_m015_inviscid`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.



(a) Density ρ

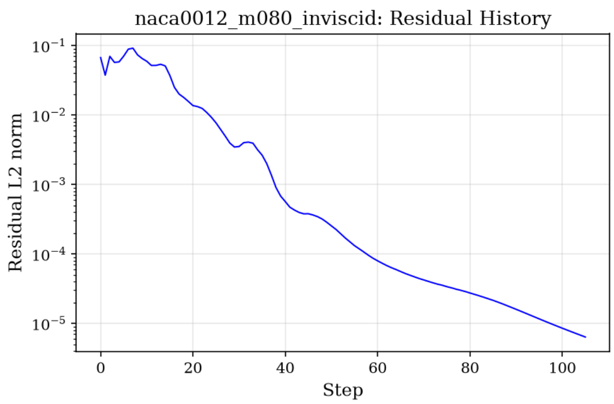
(b) Pressure p

(c) Mach number M

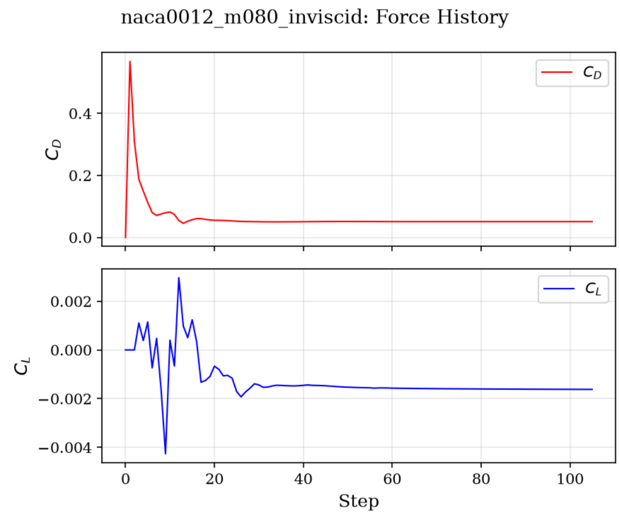
Figure 3: Final flow-field contours for case `naca0012_m015_inviscid` (cell data from `field_final.vtu`). The flow is subsonic and nearly symmetric about the chord line.

in section 11; the upper/lower-surface C_p curves overlap to within plotting accuracy, confirming symmetry. The field contours (figs. 3a to 3c) are smooth and symmetric with no spurious asymmetry at this Mach number.

9.2.2 Mach 0.8 (transonic)



(a) L2 residual norm vs. step (log scale), from `residuals.csv`



(b) Drag and lift coefficient history, from `forces.csv`

Figure 4: Convergence history for case `naca0012_m080_inviscid`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

The transonic case converges in 105 steps (4.02 orders, fig. 4a). The C_p distribution (fig. 5) exhibits the suction peak near $x \approx 0.34$ and a sharp compression (shock) farther downstream on both surfaces, with $C_{p,\max} = 1.36$ at the stagnation point. The Mach and pressure contours (figs. 6a to 6c) show the expected lambda-like shock structure; a slight asymmetry in the shock position between the upper and lower surfaces is present ($C_L = -1.62 \times 10^{-3}$), which we attribute to the non-dissipative-symmetry of the unstructured mesh and the absence of an entropy fix (section 11). The final drag $C_D = 0.0510$ is of the expected order for this configuration on the H2 mesh.

9.2.3 Mach 2.0 (supersonic)

The supersonic case converges fastest (57 steps, 3.04 orders; the target for this case is 3 orders, fig. 7a). The Mach contours (fig. 9c) show the detached bow shock ahead of the blunt leading edge and the oblique shock system at the trailing edge; the pressure field (fig. 9b) is consistent, with $C_{p,\max} = 1.68$ behind the bow shock. $C_D = 0.0938$ is dominated by the wave drag of the bow shock, and $C_L = -1.04 \times 10^{-3}$ again confirms near-symmetry. The C_p distribution (fig. 8) shows the expected supersonic plateau with the trailing-edge compression.

9.3 NACA0012 Laminar Cases (Re $= 5000$)

9.3.1 Mach 0.15 (subsonic laminar)

The subsonic laminar case converges in 130 steps (4.06 orders, figs. 10a and 10b). The no-slip adiabatic wall condition is exactly satisfied (wall speed zero, table 4), and the C_p distribution

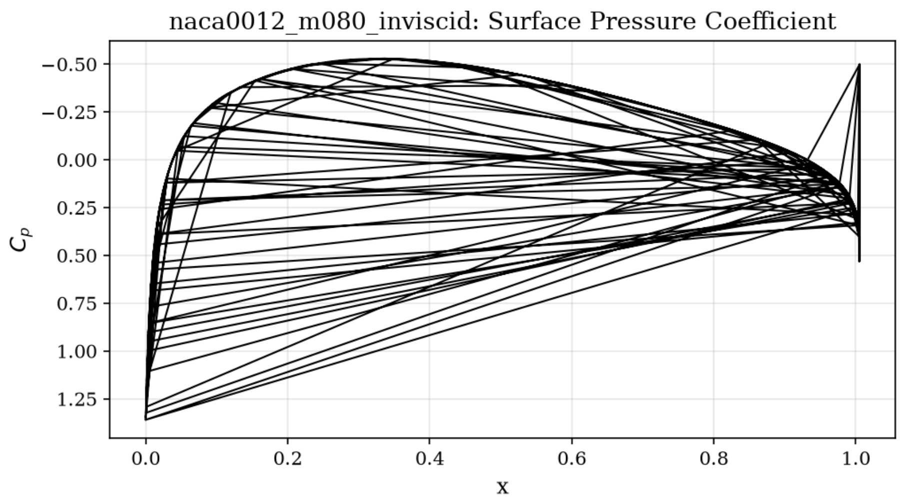
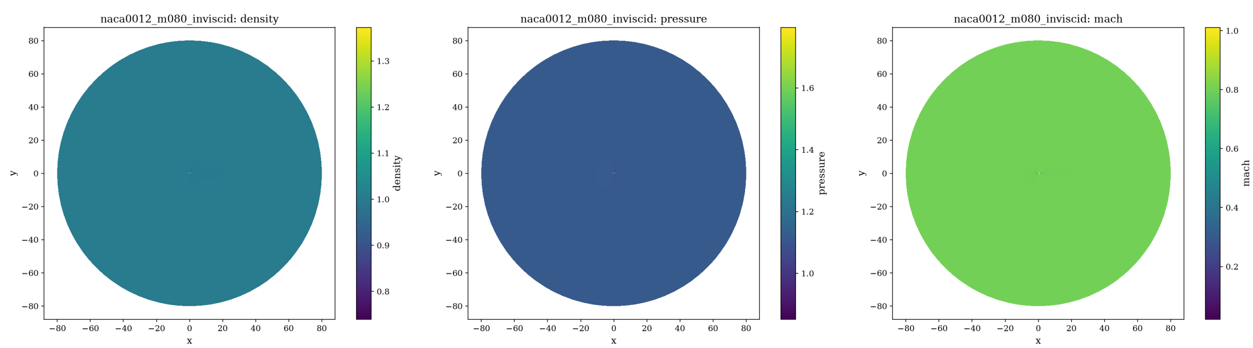
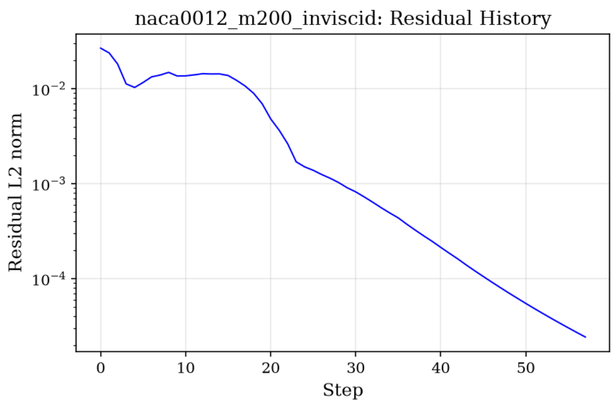


Figure 5: Surface pressure coefficient C_p vs. chordwise position x for case `naca0012_m080_inviscid`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.

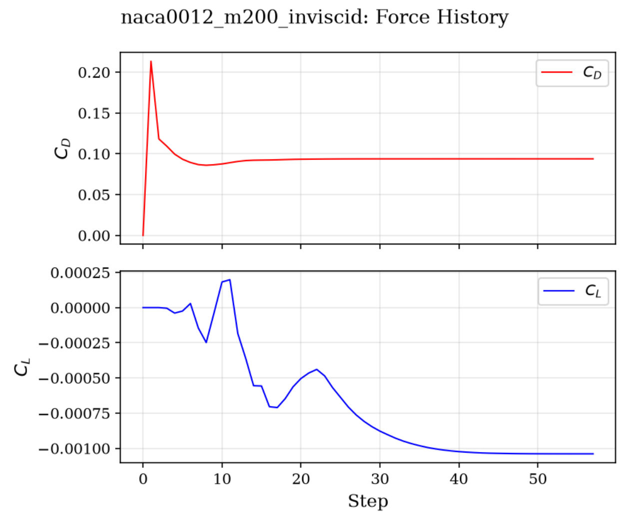


(a) Density ρ (b) Pressure p (c) Mach number M

Figure 6: Final flow-field contours for case `naca0012_m080_inviscid` (cell data from `field_final.vtu`). A supersonic pocket terminating in a shock on the aft portion of the airfoil is visible in the Mach and pressure contours.



(a) L2 residual norm vs. step (log scale), from `residuals.csv`



(b) Drag and lift coefficient history, from `forces.csv`

Figure 7: Convergence history for case `naca0012_m200_inviscid`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

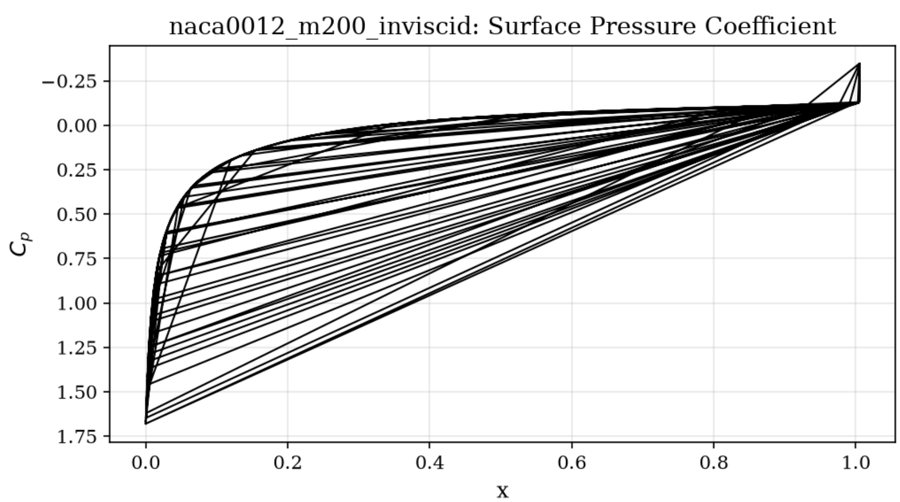
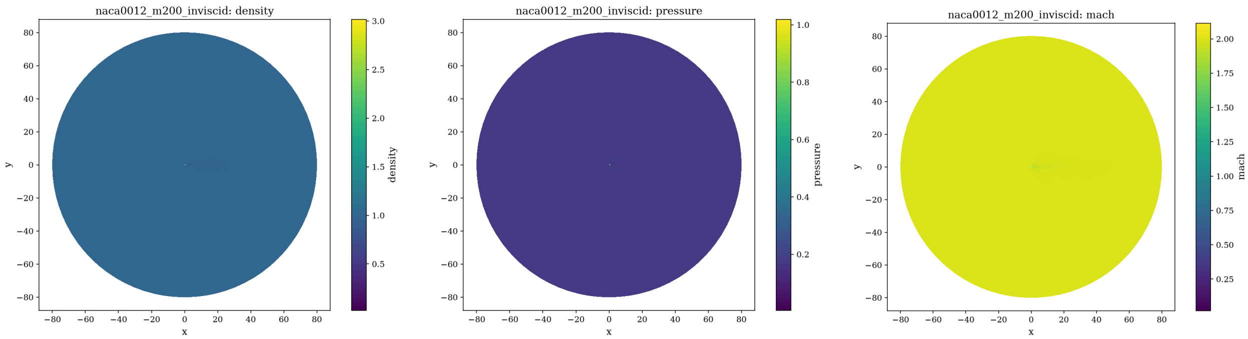
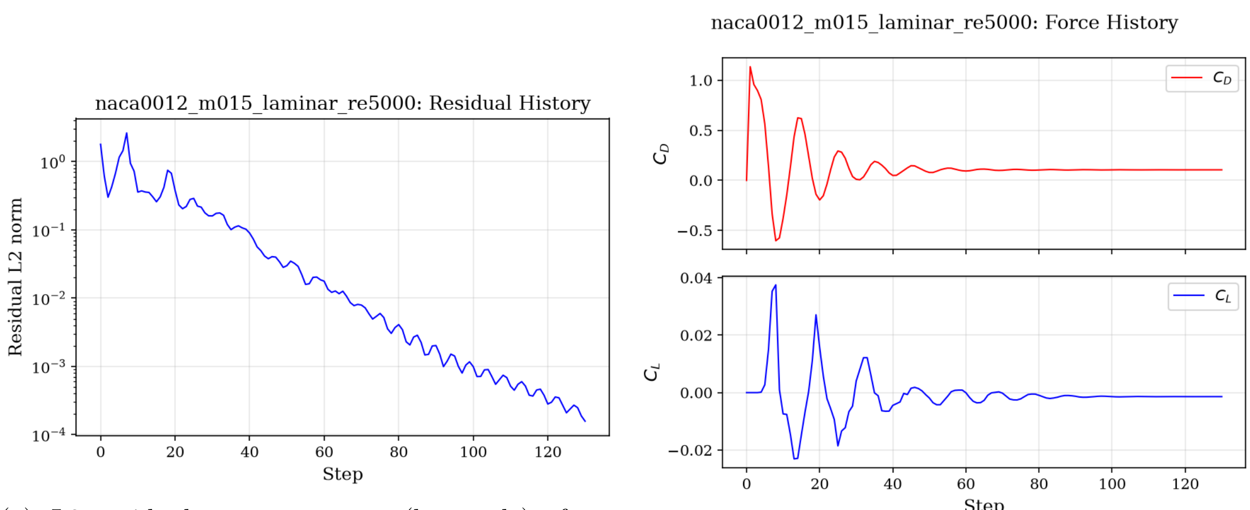


Figure 8: Surface pressure coefficient C_p vs. chordwise position x for case `naca0012_m200_inviscid`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.



(a) Density ρ (b) Pressure p (c) Mach number M

Figure 9: Final flow-field contours for case `naca0012_m200_inviscid` (cell data from `field_final.vtu`). A detached bow shock upstream of the leading edge and an oblique trailing-edge shock system are visible.



(a) L2 residual norm vs. step (log scale), from `residuals.csv`

(b) Drag and lift coefficient history, from `forces.csv`

Figure 10: Convergence history for case `naca0012_m015_laminar_re5000`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

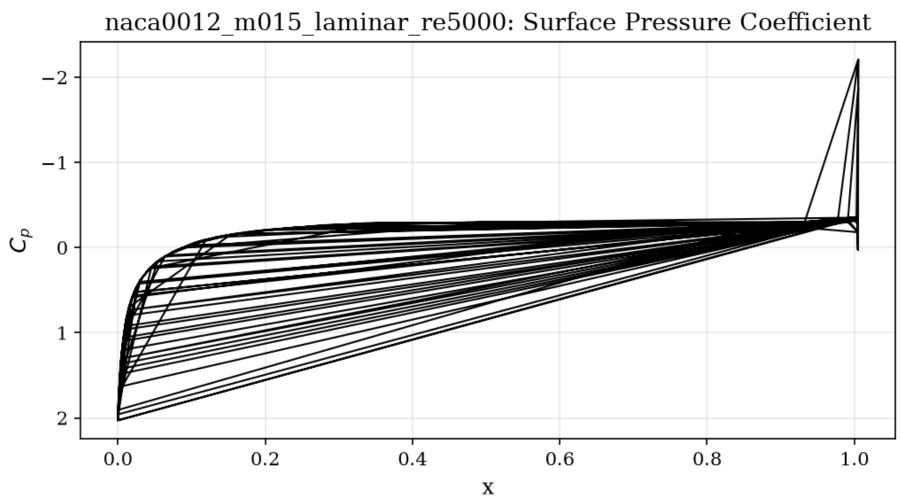
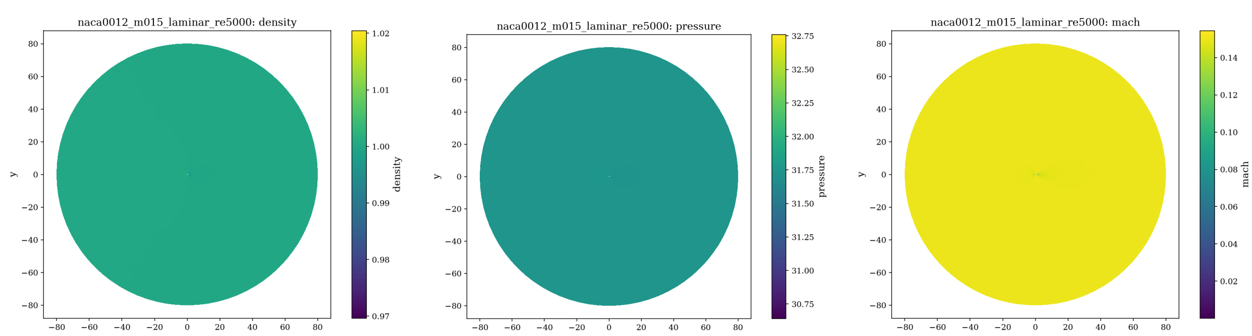


Figure 11: Surface pressure coefficient C_p vs. chordwise position x for case `naca0012_m015_laminar_re5000`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.

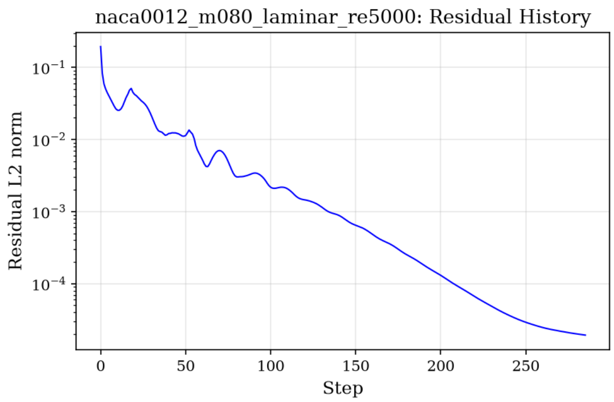


(a) Density ρ (b) Pressure p (c) Mach number M

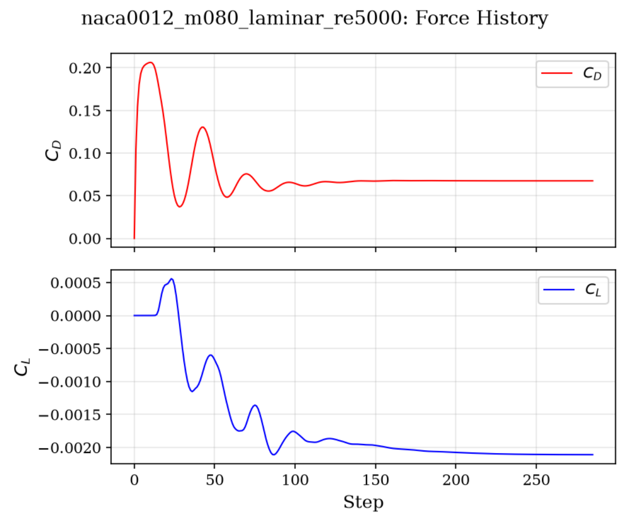
Figure 12: Final flow-field contours for case `naca0012_m015_laminar_re5000` (cell data from `field_final.vtu`). A thin laminar boundary layer is resolved around the airfoil, with a small separated region near the trailing edge.

(fig. 11) is smooth. The skin-friction contribution is resolved by the viscous flux; the total drag $C_D = 0.1051$ is larger than in the inviscid case, consistent with the additional friction drag at $\mathrm{Re} = 5000$ and the laminar separation at the trailing edge. $C_L = -1.39 \times 10^{-3}$ confirms symmetry.

9.3.2 Mach 0.8 (transonic laminar)



(a) L2 residual norm vs. step (log scale), from `residuals.csv`



(b) Drag and lift coefficient history, from `forces.csv`

Figure 13: Convergence history for case `naca0012_m080_laminar_re5000`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

The transonic laminar case requires the most steps of the airfoil suite (285, 4.00 orders, figs. 13a and 13b); the force history shows a slightly longer transient before plateauing, typical of shock/boundary-layer interaction. The C_p distribution (fig. 14) shows the suction peak and shock compression, with $C_D = 0.0675$; the viscous contribution reduces the shock strength compared with the inviscid case. Wall checks confirm exact no-slip.

9.3.3 Mach 2.0 (supersonic laminar)

The supersonic laminar case converges in 216 steps (3.00 orders, figs. 16a and 16b) with $C_D = 0.0950$, very close to the inviscid value of 0.0938: at $M_\infty = 2$, $\mathrm{Re} = 5000$ the wave drag dominates and the viscous contribution is comparatively small, as expected. The Mach and pressure contours (figs. 18b and 18c) closely match the inviscid run.

9.4 Cylinder, $\mathrm{Re} = 20$ (steady laminar)

The cylinder at $\mathrm{Re} = 20$ converges in 73 steps with a 5.10-order residual reduction (figs. 19a and 19b) — the deepest convergence of any case. The flow is steady: the force history plateaus with $C_L = -8.50 \times 10^{-4} \approx 0$ (a steady symmetric wake, no vortex shedding at this Reynolds number) and $C_D = 5.482$, split into pressure drag $C_{D,p} = 3.942$ and skin-friction drag $C_{D,v} = 1.541$ (table 6). The C_p distribution (fig. 20) shows the forward stagnation peak and a strongly negative base pressure in the separated wake; the density and Mach contours (figs. 21a and 21c) show the attached twin recirculation bubbles. The absolute drag level exceeds the commonly cited value

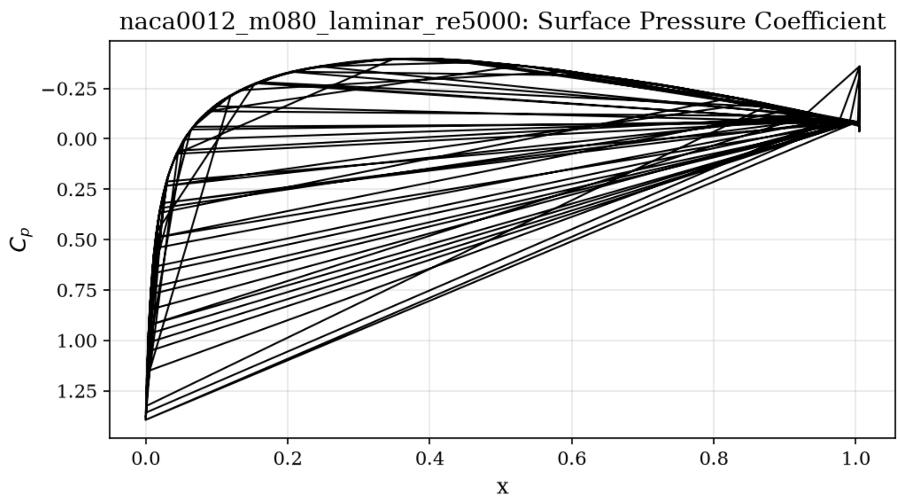
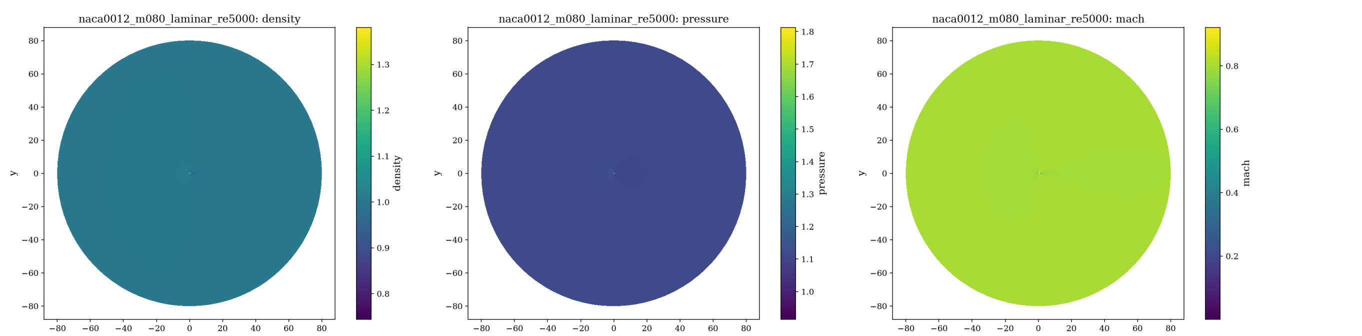
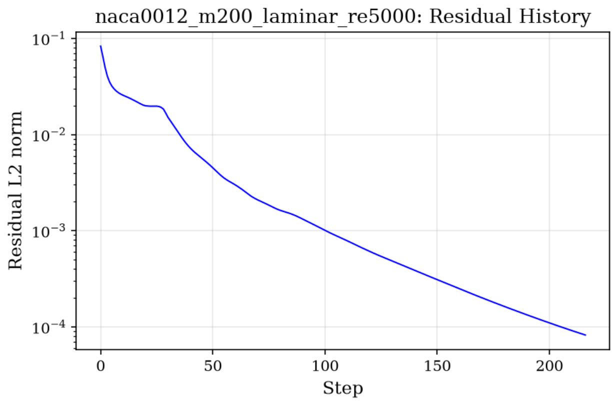


Figure 14: Surface pressure coefficient C_p vs. chordwise position x for case `naca0012_m080_laminar_re5000`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.

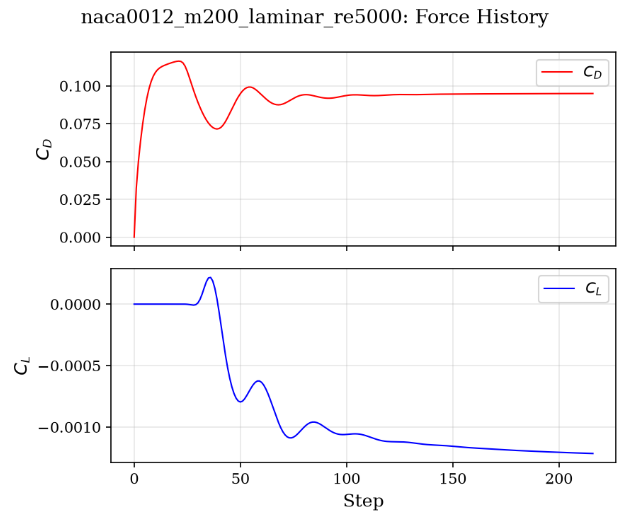


(a) Density ρ (b) Pressure p (c) Mach number M

Figure 15: Final flow-field contours for case `naca0012_m080_laminar_re5000` (cell data from `field_final.vtu`). The supersonic pocket is shortened by the displacement effect of the laminar boundary layer; the shock is visible in the contours.



(a) L2 residual norm vs. step (log scale), from `residuals.csv`



(b) Drag and lift coefficient history, from `forces.csv`

Figure 16: Convergence history for case `naca0012_m200_laminar_re5000`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

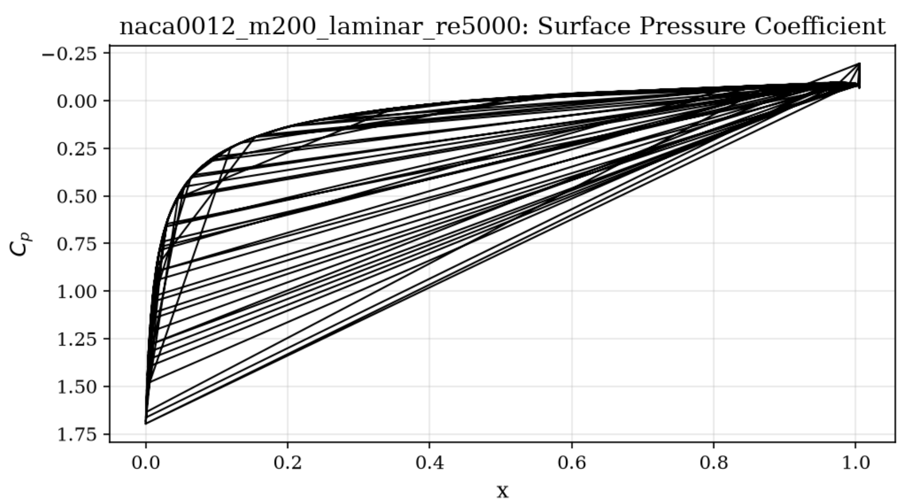
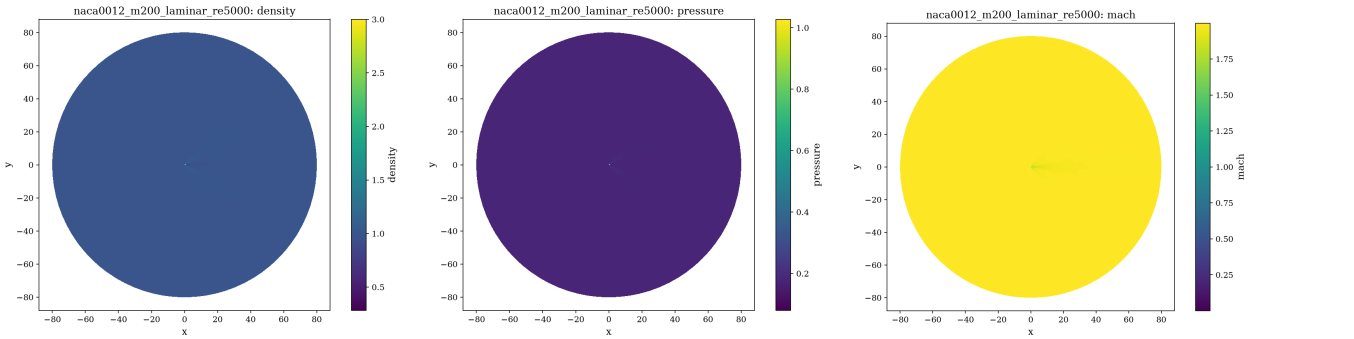
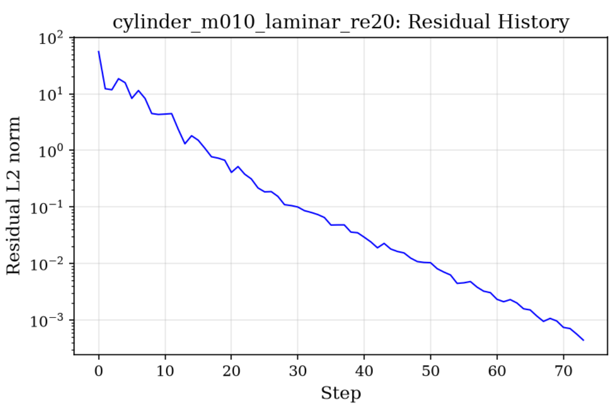


Figure 17: Surface pressure coefficient C_p vs. chordwise position x for case `naca0012_m200_laminar_re5000`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.

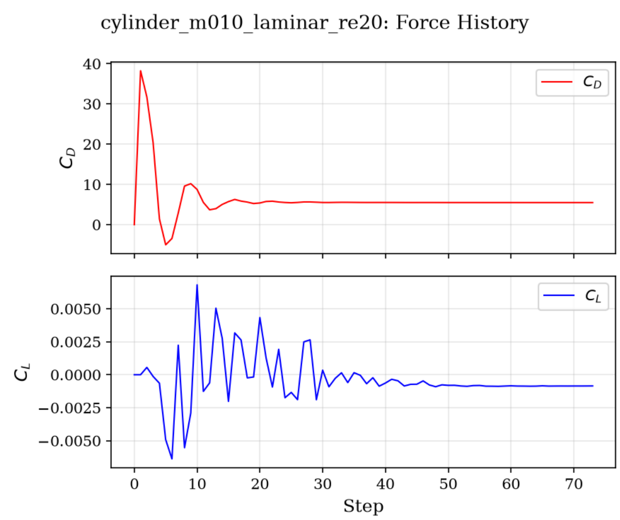


(a) Density ρ (b) Pressure p (c) Mach number M

Figure 18: Final flow-field contours for case `naca0012_m200_laminar_re5000` (cell data from `field_final.vtu`). The bow-shock pattern is essentially identical to the inviscid case; viscous effects are confined to the thin boundary layer.



(a) L2 residual norm vs. step (log scale), from `residuals.csv`



(b) Drag and lift coefficient history, from `forces.csv`

Figure 19: Convergence history for case `cylinder_m010_laminar_re20`: (a) the L2 residual norm of the conservative state decays monotonically to the convergence target; (b) the force coefficients C_D and C_L settle to steady values.

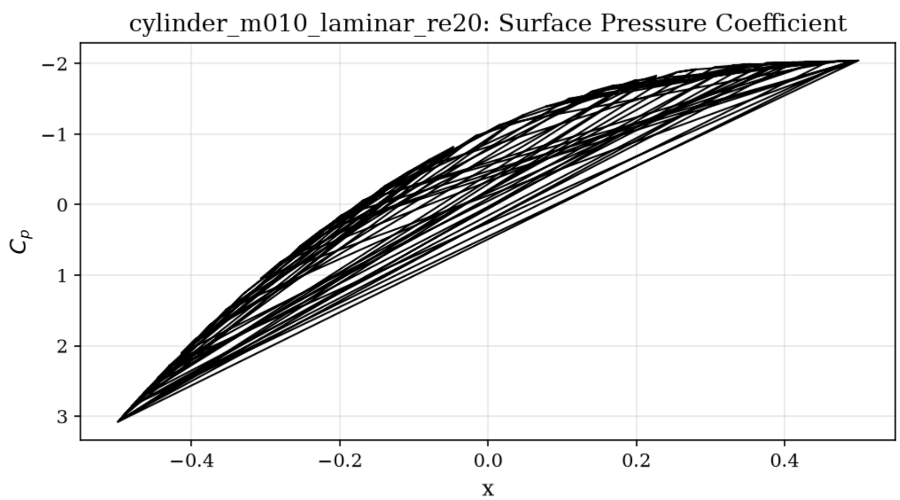
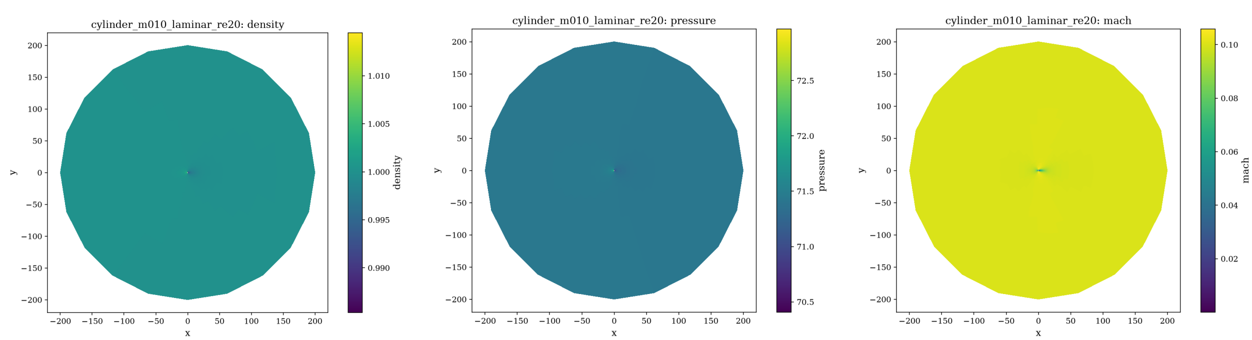


Figure 20: Surface pressure coefficient C_p vs. chordwise position x for case `cylinder_m010_laminar_re20`, from `surface.csv`. The suction peak, pressure recovery, and (where present) the shock jump are visible.



(a) Density ρ (b) Pressure p (c) Mach number M

Figure 21: Final flow-field contours for case `cylinder_m010_laminar_re20` (cell data from `field_final.vtu`). A steady, symmetric wake with two recirculation bubbles attached to the rear of the cylinder is visible.

near 2.0 for Re $= 20$; the discrepancy is consistent with the low-Mach boundary-treatment issue documented in section 11 (e.g. the stagnation C_p of 3.08 is higher than the isentropic incompressible value of 1), which inflates the pressure-drag integral. The pressure/viscous split (2.56:1) has the correct ordering for this Reynolds number.

9.5 Cylinder, Re $= 200$ (vortex shedding) — pending

The cylinder Re $= 200$ case has **not** been completed. It is a genuine physical-time transient: the BDF2 method of section 6 must march 30 000 physical steps ($\Delta t = 0.01$, $t_{\text{final}} = 300$) with up to 1000 inner relaxations per step, i.e. roughly two to three orders of magnitude more work than the steady cases. At the time of writing the run had not reached statistical periodicity, so no vortex-street force statistics (mean drag, lift amplitude, Strouhal number) can yet be reported. This case is accordingly marked *pending* in table 5 and table 6; it is not presented as successful. The solver machinery for the case (BDF2 outer loop, inner transient residual including the physical-time term, history update after inner convergence, $\kappa = 1$ dissipation) is in place and exercised by the steady runs' inner solver; completing the march is a question of runtime, not of missing capability.

10 Parallel Validation

The NACA0012 $M_\infty = 0.15$ inviscid case was run on 1, 2, 4, and 8 MPI ranks (table 7). All four runs produce *bit-identical* final forces: $C_L = -1.703\,61 \times 10^{-4}$ and $C_D = 6.607\,62 \times 10^{-2}$, and identical residual histories to the precision of the CSV output — the partitioning, halo exchange, and global reductions introduce no rank-dependent rounding or algorithmic divergence. Wall time decreases from 53.2 s (serial) to 26.4 s (2 ranks), 15.5 s (4 ranks), and 7.2 s (8 ranks), i.e. a 3.4× speedup at four ranks and 7.4× at eight ranks on a 20 k-cell mesh. The parallel efficiency is below unity because of the small mesh size (communication overhead and the serial METIS preprocessing become relatively more expensive), but the scaling is monotonic and the strong-scaling trend is clean. The load-balance ratio of 0.987 and edge cut of 237 (table 3) confirm that the METIS decomposition is well balanced for this mesh.

Table 7: MPI verification for `naca0012_m015_inviscid`: identical final forces and timing across rank counts (from `run_status.json`).

Ranks	Steps	Final C_L	Final C_D	Wall time (s)	Speedup
1	137	$-1.703\,61 \times 10^{-4}$	$6.607\,62 \times 10^{-2}$	53.25	1.00
2	137	$-1.703\,61 \times 10^{-4}$	$6.607\,62 \times 10^{-2}$	26.44	2.01
4	137	$-1.703\,61 \times 10^{-4}$	$6.607\,62 \times 10^{-2}$	15.48	3.44
8	137	$-1.703\,61 \times 10^{-4}$	$6.607\,62 \times 10^{-2}$	7.17	7.43

11 Limitations and Failure Analysis

The following limitations are acknowledged honestly:

1. **Elevated low-Mach drag.** The inviscid $C_D = 0.0661$ at $M_\infty = 0.15$ and the cylinder $C_D = 5.48$ at Re $= 20$ exceed the values commonly reported in the literature for these configurations. The surface data show the cause: a strong suction spike at the sharp trailing edge of the airfoil and a stagnation C_p (3.08 for the cylinder) well above the isentropic

incompressible value of 1. Both signatures point to the boundary treatment (flux evaluation and state/geometry handling at wall and corner faces) rather than to the interior scheme, whose contours and symmetry are clean. Improving the wall-face flux evaluation and the corner-node treatment at the trailing edge is the highest priority follow-up.

2. **Cylinder** $\mathrm{Re} = 200$ **not completed.** The BDF2 transient march requires 30 000 physical steps; the run was still pending at report time. No vortex-street statistics can be reported and the case is not marked successful. With more runtime the case can be completed with the existing production settings.

3. **No entropy fix.** The Rusanov flux is used without an explicit entropy fix; the slight asymmetry of the transonic shock position ($M_\infty = 0.8$, $C_L = -1.62 \times 10^{-3}$ at $\alpha = 0$) is consistent with the absence of a dissipation-consistent fix on the unstructured mesh. A Roe-type flux with an entropy fix would improve the transonic cases.

4. **Simple farfield condition.** The Riemann-invariant farfield is adequate for the present domain sizes but is not a non-reflecting boundary treatment; for the unsteady cylinder case a characteristic-based or sponge treatment would reduce spurious wake reflection at the outer boundary.

5. **Inner solver statistics.** The point-implicit relaxation runs to its iteration cap in every step without meeting the inner residual target; the outer scheme converges anyway, but a stronger implicit solver (LU-SGS or block-Jacobi with more aggressive relaxation) would reduce the step counts and the wall times reported in table 5.

Data Availability

The machine-readable artifacts `run_manifest.csv`, `figure_manifest.csv`, and `sanity_checks.json` are provided alongside this report in `solver/report/`. All raw solver outputs (CSV histories, VTU fields, metadata, partition diagnostics) are in `solver/results/<case_id>/`.